Uniwersytet SWPS

Task list no. 7 – In this lab, we will implement bucket sort and the randomized select algorithm.

Task 1. Implement the Bucket Sort algorithm.

1-1: Generate an array of 50 random real numbers in the range [0, 1] and print this array.

1-2: While inserting each element into its corresponding bucket, print the contents of the selected bucket immediately after the insertion (to trace how elements are distributed).

1-3: After all elements have been distributed and sorted within the buckets, print the final sorted array.

Task 2. Implement the Randomized-Select algorithm to find the i-th smallest element in an array.

2-1: Generate an array of 100 random integers in the range [10, 100] and print this array.

2-2: Execute the Randomized-Select algorithm on this array to find the 20th smallest element. During each recursive call of the algorithm, print the elements of the current subarray being partitioned, so that the partitioning process can be traced step by step.

Note. The lab report should contain your code along with the results.

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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